OMNIVISION™

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50E HDR Exposure/Gain Control

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HDR HTS

- Below is HDR mode HTS calculation.

Format	HTS
DCG	{0x380c, 0x380d}
DCG + VS	{0x380c, 0x380d} + {0x384c, 0x384d}
STG2	{0x380c, 0x380d} * 2
STG3	{0x380c, 0x380d} * 3

- Exposure register unit is HTS. One HTS time is HTS / 75MHz.
- Exposure and VTS step is 2 for binning, 4 for full size.

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SCG STG2

- Exp_L {0x3500, 0x3501, 0x3502}, Exp_M {0x3540, 0x3541, 0x3542}
- Gain_L {0x3508, 0x3509}, Gain_M {0x3548, 0x3549}
- Normal mode gain range 1 ~ 63.75x; High gain mode gain range 1 ~ 255x
- Exposure Requirement

  Host sets exposure Exp_L(n+2), Exp_M(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_L(n+2), Exp_M(n+2). Exp_L(n+2) starts in frame n+1, Exp_M(n+2) starts in frame n+2.
  - Exp_L(n+2) + Exp_M(n+1) < VTS(n+1) – 48
  - Exp_M(n+2) >= Exp_M(n+1) + Image Height – VTS(n+1), otherwise increase VTS(n+1) to meat requirement.

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SCG STG3

- Exp_L {0x3500, 0x3501, 0x3502}, Exp_M {0x3540, 0x3541, 0x3542}, Exp_S {0x3580, 0x3581, 0x3582}
- Gain_L {0x3508, 0x3509}, Gain_M {0x3548, 0x3549}, Gain_S {0x3588, 0x3589}
- Normal mode gain range 1 ~ 63.75x; High gain mode gain range 1 ~ 255x
- Exposure Requirement

Host sets exposure Exp_L(n+2), Exp_M(n+2), Exp_S(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_L(n+2), Exp_M(n+2), Exp_S(n+2). Exp_L(n+2) starts in frame n+1, Exp_M(n+2) and Exp_S(n+2) start in frame n+2.

  - Exp_L(n+2) + Exp_M(n+1) + Exp_S(n+1) < VTS(n+1) – 100
  - Exp_M(n+2) + Exp_S(n+2) >= Exp_M(n+1) + Exp_S(n+1) + Image Height – VTS(n+1), otherwise increase VTS(n+1) to meat requirement.

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DCG, DCG Combine

- Exp_DCG {0x3500, 0x3501, 0x3502}. For DCG combine set {0x3500, 0x3501, 0x3502} and {0x3540, 0x3541, 0x3542} to same value.
- Gain_HCG {0x3508, 0x3509}, Gain_LCG {0x3548, 0x3549}
- Normal mode gain range 1 ~ 15.9375x; High gain mode gain range 1 ~ 63.75x
- For DCG combine when change HCG/LCG gain, please use below setting

  6c 5006 02

  {0x3508, 0x3509} = HCG gain value

  {0x501a, 0x501b} = HCG gain value * 4

  {0x3548, 0x3549} = LCG gain value

  {0x501d, 0x501e} = LCG gain value * 4

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DCG+VS, DCGCombine+VS

- Exp_DCG {0x3500, 0x3501, 0x3502}, Exp_VS {0x3580, 0x3581, 0x3582}. For DCGCombine+VS set {0x3500, 0x3501, 0x3502} and {0x3540, 0x3541, 0x3542} to same value.
- Gain_HCG {0x3508, 0x3509}, Gain_LCG {0x3548, 0x3549}, Gain_VS {0x3588, 0x3589}
- Gain range 1 ~ 63.75x
- Exposure Requirement

Host sets exposure Exp_DCG(n+2), Exp_VS(n+2), VTS(n+1) in frame n. Frame n+2 image data is for Exp_DCG(n+2), Exp_VS(n+2). Exp_DCG(n+2) starts in frame n+1, Exp_VS(n+2) starts in frame n+2.

  - Exp_DCG(n+2) + Exp_VS(n+1) < VTS(n+1) – 48
  - No Combine: Exp_VS(n+2) >= Exp_VS(n+1) + Image Height – VTS(n+1), otherwise increase VTS(n+1) to meat requirement.
  - Combine: Exp_VS(n+2) <= VTS(n+2) - Image Height - 72

- For DCGCombine+VS when change HCG/LCG gain, please use below setting

  6c 5006 02

  {0x3508, 0x3509} = HCG gain value

  {0x501a, 0x501b} = HCG gain value * 4

  {0x3548, 0x3549} = LCG gain value

  {0x501d, 0x501e} = LCG gain value * 4

  {0x3588, 0x3589} = VS gain value

  {0x5020, 0x5021} = VS gain value * 4

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Thank You

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